DATABASE AND SQL

LEARNING OBJECTIVES

- To understand basic database queries written in Structured Query Language (SQL)
- To use the classes and interfaces of namespace System.Data to manipulate databases.
- To use the classes and interfaces of namespace System.Data.OleDb

WHAT IS A DATABASE CONNECTION?

A database connection is a communication link between your application (in this case, a VB.NET program) and a database server (such as MySQL).

Through this connection, your program can:

- Send queries (like SELECT, INSERT, UPDATE, DELETE) to the database.
- Receive results from those queries (like lists of users or records).
- Modify or store data permanently in the database.

STEPS TO USE A DATABASE

INSTALL MYSQL CONNECTOR FOR .NET

Go to the official MySQL site:
🔗 https://dev.mysql.com/downloads/connector/net/

Download and install the MySQL Connector/NET that matches your system and .NET version.

After installation, the connector will automatically add the MySQL.Data.dll library to your GAC (Global Assembly Cache).

PREPARE XAMPP AND MYSQL

1. Open XAMPP Control Panel.
2. Start Apache and MySQL (both show green).
3. Click Admin next to MySQL to open phpMyAdmin (or go to http://localhost/phpmyadmin).

CREATE DATABASE AND TABLE (PHPMYADMIN)

ADD REFERENCE IN VISUAL STUDIO

1. Open your VB.NET project in Visual Studio.
2. Go to Project → Add Reference → Assemblies → Extensions.
3. Find and check MySql.Data.
4. Click OK.

ADD MYSQL CONNECTOR/NET TO YOUR VB.NET PROJECT

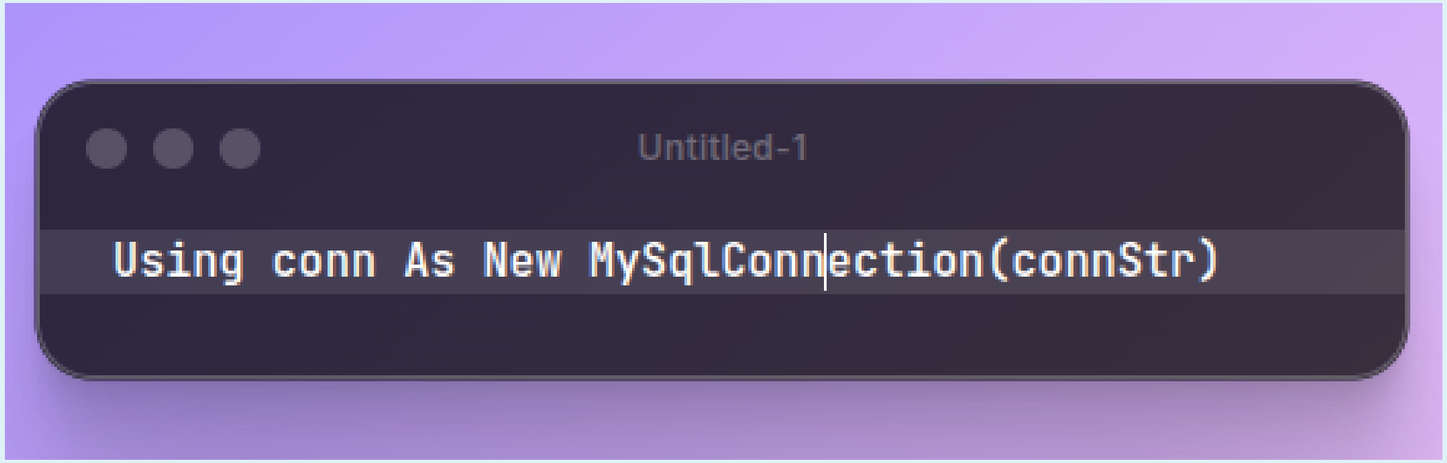
Download + install from MySQL site; then add reference to MySql.Data.dll in your project (Project → Add Reference).

```vbnet
Imports MySql.Data.MySqlClient
```

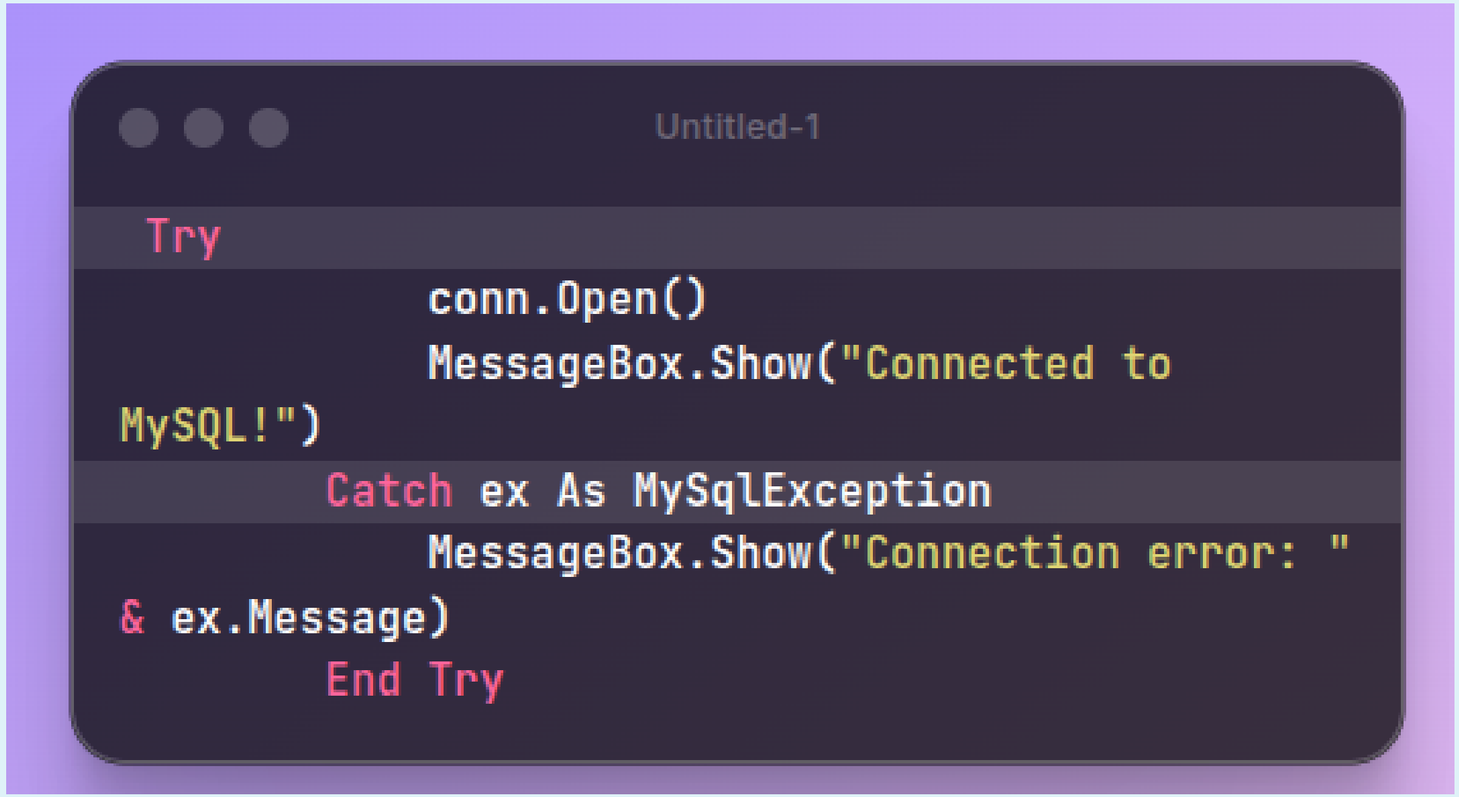
BUILD A SAFE CONNECTION STRING



USE USING BLOCKS FOR CONNECTION & COMMANDS (RECOMMENDED PATTERN)



USE USING BLOCKS FOR CONNECTION & COMMANDS (RECOMMENDED PATTERN)



Untitled-1

```
Private Sub TestConnection()
    Dim connStr As String =
"server=localhost;user
id=root;password=;database=testdb;port=3306;"
    Using conn As New MySqlConnection(connStr)
        Try
            conn.Open()
            MessageBox.Show("Connected to
MySQL!")
        Catch ex As MySqlException
            MessageBox.Show("Connection error: "
& ex.Message)
        End Try
    End Using
End Sub
```

```vb
Imports MySql.Data.MySqlClient

Public Class Form1
    Private Sub btnLogin_Click(sender As Object, e As EventArgs) Handles btnLogin.Click
        ' --- Connection string ---
        Dim connStr As String = "server=localhost;user id=root;password=;database=login_db;"
        Dim conn As New MySqlConnection(connStr)

        Try
            conn.Open()

            ' --- Query to check if username & password exist ---
            Dim query As String = "SELECT * FROM users WHERE username=@username AND password=@password"

            Dim cmd As New MySqlCommand(query, conn)
            cmd.Parameters.AddWithValue("@username", txtUser.Text)
            cmd.Parameters.AddWithValue("@password", txtPass.Text)

            Dim reader As MySqlDataReader = cmd.ExecuteReader()

            If reader.HasRows Then
                MessageBox.Show("Login Successful!", "Welcome", MessageBoxButtons.OK, MessageBoxIcon.Information)
                ' Optionally open another form here
                ' Dim dashboard As New Form2()
                ' dashboard.Show()
                ' Me.Hide()
            Else
                MessageBox.Show("Invalid username or password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error)
            End If

            reader.Close()
            conn.Close()

        Catch ex As MySqlException
            MessageBox.Show("Database Error: " & ex.Message)
        End Try
    End Sub
End Class
```